

Abdul Khurram

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EDUCATION & AWARDS

University of British Columbia

Vancouver, BC, Canada

Bachelor of Science in Computer Science (Expected Graduation: May 2027)

Euclid Mathematics Contest, University of Waterloo

2021

Top 5% Internationally, Certificate of Distinction

EXPERIENCE

Software Developer Co-op | *Apache Spark, Python, Airflow, SQL, Linux*

Jan. 2026 – Present

RBC Royal Bank

Vancouver, BC, Canada

- Engineered **Python** pipelines orchestrated by **Airflow DAGs** to process **50M+ daily transactions** from **Delta Lake**, feeding downstream **ML models** and analytics on **petabyte-scale** financial data.
- Optimized **Spark** jobs by refactoring **SQL** queries and tuning **AQE/cluster memory**, solving resource bottlenecks and reducing **latency** by **40%**.
- Developing **multi-region** data pipelines with built-in **fault tolerance**, **idempotent** re-runs, and automated **failover** for critical banking workloads.
- Implementing job **monitoring** and **alerting** to track **SLAs** and ensure pipeline reliability, enabling rapid incident response for downstream teams.

Software Engineer Intern | *TypeScript/Node.js, AWS, Playwright, OpenAI*

Jan. – Dec. 2025

Contello.ai

Toronto, ON, Canada

- Engineered real-time **ETL** pipelines on **AWS (SQS/S3)** with **TypeScript/Next.js** + **Node** dashboards; owned design→deploy→diagnostics/**DevOps** (**CloudWatch**) at **thousands/day** scale.
- Accelerated content prep by ingesting brand docs with an **NLP** step, caching embeddings, and auto-injecting channel-specific context while enforcing **API guardrails**.
- Raised scheduler reliability to **99.9%** by moving generation to **OpenAI Responses API**, integrating **Playwright E2E** gates into **CI/CD**, hardening auth/session, and adding **retry/backoff** plus **CloudWatch alerting**.

Software Developer & Team Lead | *Unreal Engine, C++, Python*

Sept. – Dec. 2024

UBC Emerging Media Lab

Vancouver, BC, Canada

- Led an **AI-powered** VR platform in **Unreal (C++)** with interactive UI **state machines** for clinical scenarios.
- Increased stability with a **multithreaded supervisory layer** (**C++/Python**) that monitors the AI patient and auto-corrects off-protocol replies.
- Delivered a functional prototype in two weeks, securing project funding.
- Cut backend costs by **90%** via system optimizations, improving real-time performance by **40%**.

PROJECTS

SoloPilot | *Python, TypeScript/Next.js, AWS, LangChain*

Personal Project

- Building a client-to-codebase bridge for solo freelancers that unifies client intake, planning, proposals, and delivery in one workspace.
- Shipped an email intake console that triages threads, captures scope, and renders versioned proposal PDFs with freeform annotations via a **Vision Model** that captures and translates intent for precise edits.
- Built a **Python** evaluator→revisor **agent** loop with rubric-based **benchmarking** (score + revision), improving response quality via measured iterations.

Backer | *TypeScript/Next.js, AWS DynamoDB, NextAuth*

Hack The Coast 2026

- Trained an **XGBoost** **ranker** for a TikTok-style investor feed, scoring startups using **6 features** (stage affinity, text similarity, tag overlap, popularity) with **sigmoid scoring** and exploration noise.
- Built a **supervised training pipeline** ingesting like/commit positives and sampled negatives from **DynamoDB**, with online enrichment from impression and watch-milestone **event streams**.
- Implemented the **ML feature layer**: investor affinity contexts, token-level text matching, and popularity normalization.

TECHNICAL SKILLS

Languages: TypeScript/JavaScript, Java, Python, SQL, C/C++, Swift, HTML/CSS, React Native

Platforms & Tools: Next.js, Node.js/Express, Playwright; AWS (Amplify, Lambda, DynamoDB, S3, SQS, CloudWatch); OpenAI API; Meta/LinkedIn/X/Canva APIs; Git, Docker, Yarn, Tailwind CSS, RDBMS